

Ideation Phase

Empathize & Discover

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is used to understand a user's behaviours, attitudes, goals, frustrations, and expectations from the user's own point of view.

For the Metro Ticket Generating System, the primary persona is a metro passenger who wants a quick, accurate, and convenient way to book a journey and receive a verifiable digital ticket. A secondary operational persona is also considered because station staff and metro operations teams must validate and manage the ticketing process.

Primary User Persona

Persona	Metro Passenger / Daily Commuter
Primary Goal	Book a metro ticket quickly, know the correct fare, and obtain a digital ticket that can be verified easily.
Context	Travelling between metro stations and using a digital self-service booking experience.
Needs	Simple form, correct fare, fast confirmation, request visibility, and reliable QR ticket.
Main Frustrations	Queues, manual processes, fare uncertainty, paper dependency, slow validation, and lack of centralized booking status.

Empathy Goal

Understand what a passenger experiences before, during, and after metro ticket booking so that the ServiceNow solution can reduce unnecessary effort, improve trust in fare and ticket information, and provide a smoother digital journey.

Empathy Map - Metro Passenger / Commuter

THINK & FEEL

- I want to complete my booking without waiting in a long queue.
- I need confidence that the fare shown is correct.
- I expect the ticket to be available immediately after booking.
- I want a simple digital process that does not require technical knowledge.
- I need to know that my ticket can be verified at the station.
- I do not want to worry about losing a paper ticket.

SEE

- Crowded counters during peak travel hours.
- Passengers waiting for physical tickets.
- Multiple fare and route options that may be confusing.
- Other commuters using mobile phones for travel-related services.
- Digital payment and QR-based systems becoming common in public services.
- Station staff checking tickets before entry or travel.

- I expect the system to clearly show whether my booking was successful.

HEAR

- Other passengers discussing long queues or waiting time.
- Announcements about metro routes, stations, and service changes.
- Friends or commuters recommending digital booking methods.
- Station staff asking passengers to keep tickets ready for validation.
- General expectations that public transport should provide faster digital services.

SAY & DO

- Selects the starting station and the destination station (Starting From / Going To).
- Chooses the Type of Journey (single or return) and the Number of Passengers.
- Checks the fare amount displayed automatically on the form before submitting the request.
- Selects the payment mode.
- Submits the booking through the ServiceNow portal.
- Checks the request status if confirmation is not immediate.
- Opens the generated QR ticket for verification.
- Shows the digital ticket to station staff when required.

PAIN

- Time lost in queues or manual booking processes.
- Uncertainty when the calculated fare does not match expectations.
- Accidentally choosing the same station as source and destination, or leaving booking details incomplete.
- Difficulty if booking information is missing or not stored correctly.
- Dependence on paper-based confirmation.
- Slow or inconvenient ticket verification.
- Lack of clear visibility into request status.
- Frustration when a digital interface behaves differently across portal views.

GAIN

- Faster self-service ticket booking.
- Automatic and consistent fare calculation, instantly updated when stations, passengers or journey type change.
- Only the relevant amount field shown for the selected journey type.
- Instant digital ticket availability.
- QR-based ticket that is easy to present and verify.
- Centralized storage of booking information.
- Ability to track booking/request status.
- Reduced paper usage and improved overall convenience.

Secondary Empathy Map - Station Manager / Metro Operations

Although the passenger is the primary customer, the solution must also support the operational users who verify, monitor, and maintain ticketing information. Their perspective influences workflow automation, data accuracy, access control, and maintainability.

THINK & FEEL	SEE
• Ticket data should be accurate and easy to verify. • The workflow should reduce repeated manual work. • Errors should be easy to trace and correct. • The system should be maintainable by ServiceNow administrators. • Access to passenger and booking data should be controlled properly.	• Large numbers of passengers during peak periods. • Requests arriving from the Service Catalog. • Station records in the Metro Station's Details table and ticket records in the custom Metro Database table. • Flow Designer executions and request statuses. • QR tickets presented by commuters.
HEAR	SAY & DO
• Passengers asking why fares or ticket information are incorrect. • Operations teams requesting reliable booking data. • Administrators asking for simpler, maintainable workflows. • Station staff reporting validation or display issues.	• Checks request and ticket information. • Uses Flow Designer Execution Details to investigate failures. • Verifies data stored in the Metro Database and the station table. • Validates role-based access and ACL behaviour. • Tests fare, UI policy, ticket and QR behaviour in the portal.
PAIN	GAIN
• Fare mismatches. • Missing or misaligned variable-to-field mappings. • QR rendering inconsistencies in Service Portal. • Flow failures that are difficult to trace without execution logs. • Too much custom scripting increasing maintenance effort.	• Centralized and structured booking data. • Low-code workflow automation through Flow Designer. • Reliable QR-based ticket validation. • Easier troubleshooting through execution details. • Maintainable configuration with minimal custom scripting.

Empathy Insights Converted into Design Decisions

Empathy Insight	Design Response
Passenger wants less waiting	Use a self-service Service Catalog booking form.
Passenger wants fare confidence	Automate fare calculation from station order, passenger count and journey type, and auto-fill the amount fields on screen.
Passenger wants a clean, uncluttered form	Use UI policies to show only the amount field relevant to the selected Type of Journey.
Passenger should not make invalid bookings	Block the same station as Starting From and Going To through catalog client script validation.
Passenger wants proof of booking	Generate a QR-based digital ticket.
Passenger wants visibility	Provide request status tracking.
Operations need accurate records	Store station data and booking data in structured custom tables inside the scoped application.
Operations need automation	Use Flow Designer for processing and task creation.
Admins need maintainability	Keep client scripts and custom logic minimal where possible.
Station staff need reliable validation	Use a Service Portal-compatible widget and spModal approach for QR display.

Final Empathy Statement

The metro passenger needs a fast, understandable, and trustworthy digital ticketing experience with accurate fare calculation, clear booking confirmation, and an easily verifiable QR ticket. At the same time, metro operations need centralized data, dependable workflow automation, secure access, and easy troubleshooting. These combined needs directly shaped the ServiceNow solution architecture.